Curated to Win: A Reward-Model Leaderboard's Lead Is Only as Robust as Its Score Gap

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Abstract

Automated research and self-improvement pipelines pick reward models and LLM judges by their accuracy on a validation set, and that set is often curated by the same pipeline or its designers. We ask how much the choice of items can move the winner. For any benchmark score, we prove that making a trailing model overtake the leader requires removing a share of the benchmark at least equal to the score gap between them. On RewardBench (118 models, 2985 items) and RewardBench 2 (188 models, 1753 items), linear programs show that curation reaches this floor at the top of the leaderboard (median cost 1.00× the gap). Every model in RewardBench's top 20 can be placed first by keeping at least 89.5% of the benchmark. The runner-up takes first place when 19 items (0.7% of the score weight) are removed. Keeping every section's share fixed raises the cost only modestly. Curation need not be adversarial to matter: on a random 100-item validation set, RewardBench's official leader comes first in only 35% of draws. We recommend reporting a lead as a curation margin, and fixing validation items before the pipeline sees any results.

1 Introduction

Systems that automate research or improve themselves need an evaluator: a reward model to optimise against, or an LLM judge to rank candidate outputs (Lu et al., 2024; Zheng et al., 2023). Evaluators are chosen by meta-evaluation, their agreement with human preference labels on a benchmark such as RewardBench (Lambert et al., 2024; Malik et al., 2025). In an automated pipeline, the items used for that choice are often selected by the pipeline itself: a filtered subset, a domain slice, or a quick validation set. If the ranking of evaluators depends on which items are kept, then whoever keeps the items also chooses the evaluator. This is a form of reward hacking one level above the reward (Gao et al., 2022).

Earlier work shows that leaderboards are sensitive to benchmark choice, answer format and private testing (Dehghani et al., 2021; Alzahrani et al., 2024; Singh et al., 2025; Zhang and Hardt, 2024), and to targeted edits of items (Dubrovsky et al., 2025). Two recent analyses are closest to ours. Gordienko et al. (2026) count the *datasets* a developer must train on to top a multi-task leaderboard and show the problem is NP-hard under mean win rate; Oyarhoseini et al. (2026) use influence approximations to show that sub-1% match-level edits can change the leader of Bradley–Terry arenas. We study a different operation, removing individual *evaluation items*, for the additive accuracy scores that reward-model and judge meta-evaluation use. There the question is a linear program, it has an exact closed-form lower bound, and we can measure how close curation comes to it. We make three contributions:

- A floor: overtaking the leader requires removing at least the score gap (Prop. 1).
- A measurement: linear programs over item weights, rounded to real subsets and rechecked, show that curation reaches the floor near the top of RewardBench and RewardBench 2, and stays within about twice it when each section keeps its share.
- A benign baseline: random validation sets of realistic size often crown a model other than the official leader.

2 Setup

A benchmark has items $i = 1, \dots, N$ with official weights $w_i^0 \geq 0$, $\sum_i w_i^0 = 1$. Model m has per-item score $r_{mi} \in [0, 1]$ and official score $S_m = \sum_i w_i^0 r_{mi}$. A *curation* keeps a set K with kept share $f = \sum_{i \in K} w_i^0$ and rescores each model on it, $S_m^K = \sum_{i \in K} w_i^0 r_{mi} / f$. We call $1 - f$ the *removed share*.

Data. RewardBench v1 publishes per-item results for 118 models on 2985 preference pairs in 23 subsets (Lambert et al., 2024). We rebuild each official score from these results: section scores are count-weighted means of subset accuracy, with `math-prm` upweighted as in the released code, and the overall score is the mean of the four sections. For four spot-checked models, from the leader to a 0.5B chat model, our subset accuracies equal RewardBench's published per-subset scores exactly. Result files are aligned by position, and a text check found no misaligned items. RewardBench 2 (Malik et al., 2025) publishes results by item id. We use the 188 models with results on all 1753 items of its five accuracy subsets, scored as the mean of subset accuracies. We leave out the Ties subset, which is scored differently.

Crowning. Model m is *crowned* by K if $S_m^K > S_k^K$ for every other model k. For a fixed f, crowning is a linear feasibility problem in relaxed weights w:

$$\max_{w,t} \; t \;\text{ s.t. }\; \textstyle\sum_i w_i(r_{mi} - r_{ki}) \geq t \;\; \forall k \neq m,$$
$$0 \leq w_i \leq w_i^0/f, \quad \textstyle\sum_i w_i = 1,$$

and m is crownable at f if $t^* > 0$. A *section-preserving* variant also fixes each section's total weight at its official value, so the curated set keeps the benchmark's composition. We scan f downward from 1 and then bisect. Each LP solution is rounded to a real subset: items at or near their upper bound are kept, and each keeps its official weight, renormalised. The subset is rescored, and it counts only if m is still strictly first. All reported shares are those of the rounded subsets. Both searches are heuristic in f, and a verified section-preserving subset is also a valid unconstrained one, so the unconstrained cost we report is the smaller of the two. For averaged scores each feasibility check is a linear program of size $N \times M$; this contrasts with ordinal aggregates such as mean win rate, where the analogous selection problem is NP-hard (Gordienko et al., 2026).

3 A Floor on the Cost of Curation

Proposition 1. *If $S_a - S_b = g > 0$, then every K with $S_b^K \geq S_a^K$ removes a share of at least g.*

Proof. Let $D_i = r_{ai} - r_{bi} \in [-1, 1]$. Then $f\,(S_a^K - S_b^K) = \sum_{i \in K} w_i^0 D_i = g - \sum_{i \notin K} w_i^0 D_i \geq g - (1 - f)$, which is positive whenever $1 - f < g$. □

Crowning a model means overtaking the leader, so the removed share is at least the model's gap to the leader. The bound is tight pairwise under a simple condition. Let $W_{ab} = \sum_i w_i^0 \mathbf{1}[r_{ai} = 1, r_{bi} = 0]$ be the weight of items a gets right and b gets wrong. If $W_{ab} \geq g$, removing weight $g + \epsilon$ from those items makes b overtake a for any $\epsilon > 0$, so the pairwise cost is exactly g in the limit, up to the granularity of one item. On RewardBench this condition holds for 113 of the 117 trailing models against the leader. With many competitors it is not obvious that the floor is reachable: removing items to help m against the leader can let a third model through. Whether it is reachable is an empirical question, which we answer next.

4 Results

The floor is reached near the top. On RewardBench, all 118 models can be crowned. The median ratio of removed share to score gap is 1.00 for unconstrained curation, and at most 1.02 in the top 20, so near the top the multi-model cost is the pairwise floor (Figure 1, Table 1). The worst case over all models is 2.00. The leader's margin over the runner-up is 0.7 points. Removing 19 of 2985 items (0.7% of the score weight) makes the runner-up the winner. Every model in the top 20, down to one 10.5 points behind the leader, can be crowned while keeping at least 89.5% of the benchmark. In total, 19 of 118 models can be crowned keeping 90%, and 58 keeping 80%. RewardBench 2 behaves the same way: the median ratio is 1.00 and the maximum is 1.35. Its runner-up needs 5.3% removed against a 5.2-point gap.

Preserving composition raises the cost modestly. A curator who keeps every section at its official share leaves no trace in the section breakdown. On RewardBench this costs a median 1.39 times the gap and at most 2.31, and every top-20 model can still be crowned keeping at least 81.8%. On RewardBench 2 the median is 1.12 and the maximum 1.58.

Benign curation. Table 2 draws random validation sets of n items, with items drawn in proportion to their official weight so that a set's mean score is unbiased for the official score. With $n = 100$, RewardBench's leader, 0.7 points clear, comes first in 35% of draws, and 20 different models win at least once. RewardBench 2's leader is 5.2 points clear and keeps first place in 87% of

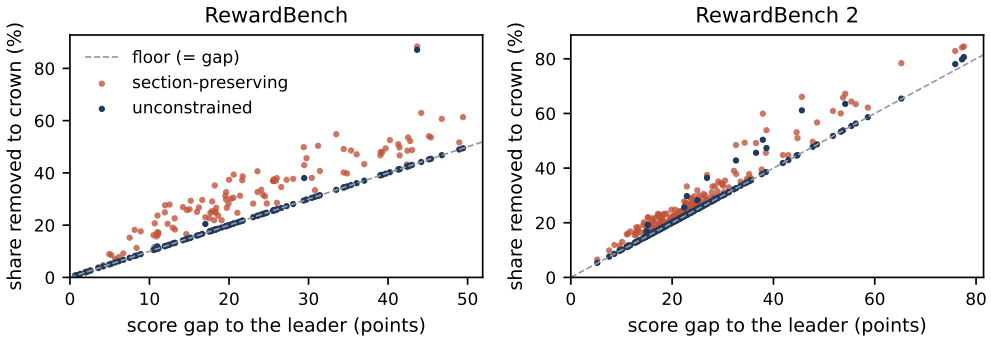


Figure 1: Share of the benchmark that must be removed to put each model first, against its official gap to the leader. Every point sits on or above the dashed floor (Prop. 1). Unconstrained curation reaches the floor for nearly every model; keeping each section's share costs more, but at most 2.31 times the gap.

			Removed to crown		
Model	Score	Gap	Share	Items	Strat.
QRM-Gemma-2-27B	94.4	0.67	0.68	19	0.72
Skywork-Reward-Gemma-2-27B-v0.2	94.3	0.84	0.85	30	0.85
Skywork-Reward-Gemma-2-27B	93.8	1.30	1.33	38	1.34
QRM-Llama3.1-8B-v2	93.1	1.97	1.98	61	1.98
Skywork-Reward-Llama-3.1-8B-v0.2	93.1	1.98	2.02	59	2.06
URM-LLaMa-3.1-8B	92.9	2.17	2.19	71	2.19
Skywork-Reward-Llama-3.1-8B	92.5	2.58	2.58	80	2.77

Table 1: The seven models behind RewardBench's leader: official score and gap to the leader (points), the share of score weight and number of items whose removal crowns each model, and the share needed when every section keeps its official weight (Strat.).

	RewardBench		RewardBench 2	
n	leader kept	winners	leader kept	winners
25	14%	60	41%	113
50	25%	33	66%	64
100	35%	20	87%	18
200	48%	12	97%	5
500	65%	9	100%	2
1000	78%	6	100%	1

Table 2: Random validation sets of n items (2,000 draws each). "Leader kept": share of draws where the official leader is strictly first after random tie-breaking. "Winners": number of distinct models that ever come first.

100-item draws: as Prop. 1 suggests, the size of the lead, not the benchmark, sets how robust the ranking is. Natural slices also move the winner. Scoring on a single section produces 3 different winners across the four sections. Among slices with a unique winner, the donotanswer slice alone is won by the model ranked #95 overall.

5 Implications for Automated Pipelines

A pipeline that selects its evaluator on items it curated, or on a slice it chose after seeing results, gains nothing from a lead smaller than the share of items it could plausibly have dropped. Three practices follow. (i) Report the *curation margin* of a selection: by Prop. 1 it is at least the score gap, and near the top of both leaderboards it equals it, so RewardBench's 0.7-point lead is a 0.7% curation margin. (ii) Fix and hash validation items before any candidate is scored, and log every subset or filter the pipeline applies. (iii) Size validation sets for rank stability, not only for accuracy. A hundred well-chosen items can estimate a model's accuracy closely (Maia Polo et al., 2024), but when the lead is under a point, as on RewardBench, the top-1 choice at $n \leq 200$ is close to a lottery (Table 2), consistent with measured benchmark variance (Madaan et al., 2024).

Limitations

We study one curation operation, removing items, on two reward-model benchmarks. Reweighting within the kept set, paraphrasing items or changing the prompt format would give a curator more room, so our costs are upper bounds on what curation needs. The LP relaxation is rounded and reverified. The reported shares are therefore achievable, but they may not be the smallest possible. A model exactly tied with a variant on every item could never be strictly first; no such pair occurred, and every model on both benchmarks was crownable. We do not claim that any published result was curated. RewardBench 2's Ties subset is excluded, so its scores differ slightly from the official ones.

Reporting and Reproducibility

All results come from public per-item result files (`allenai/reward-bench-results`, `allenai/reward-bench-2-results`). All analysis ran on one laptop CPU with SciPy's HiGHS solver, and each full sweep takes one to two hours. No models were run. One failed run is worth reporting: the first version of the section-preserving search began its bisection at a very small kept share, where rounding can empty a section, and it wrongly reported the leader as not crownable. The fixed search scans from $f = 1$ downward. Code and derived data will be released with the camera-ready version.

Use of AI Assistants

An AI assistant (Claude, Anthropic) helped write the analysis code and draft the text. The author chose the study, checked the proof and the results, and takes responsibility for the content.

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